Two-Ray Multipath Propagation & Breakpoint Distance

Electromagnetic Field Derivation, Asymptotic d^{-4} Law, and Ray-Tracing Engine Validation

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Physical Propagation: The Failure of Free-Space Friis

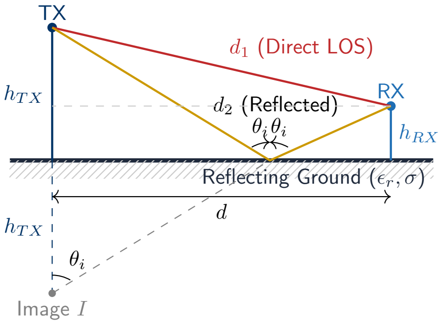
The Free-Space (Friis) Model:

- Assumes unbounded free space: $P_{RX} \propto d^{-2}$.
- In terrestrial or street-canyon links, boundary reflections (ground or lateral walls) are always present.
- Friis severely overestimates range (predicting hundreds of kilometers for cellular base stations).

Two-Ray Interference

The superposition of a direct wave (LOS) and a single specularly reflected wave creates an interference pattern that transitions from an oscillating d^{-2} regime into a steep d^{-4} power falloff.

Two-Ray Geometry & Image Antenna Principle



Exact Geometric Path Lengths:

$$d_1 = \sqrt{d^2 + (h_{TX} - h_{RX})^2}$$
$$d_2 = \sqrt{d^2 + (h_{TX} + h_{RX})^2}$$
$$\theta_i = \frac{\pi}{2} - \arctan\left(\frac{h_{TX} + h_{RX}}{d}\right)$$

Hypotheses:

- 2D propagation in the xy-plane.
- Linearly polarized electric field normal to the propagation plane ($\vec{E} = E\vec{1}_z$).
- Omnidirectional antennas in the plane:

$$\vec{h}_{e\perp}^{TX} = h_{e\perp}^{TX}\vec{1}_z, \quad \vec{h}_{e\perp}^{RX} = h_{e\perp}^{RX}\vec{1}_z$$

Far-Field Radiation & Induced Open-Circuit Voltage

The far-field electric field radiated by the transmitter driven by input current I_a^{TX} at distance r is:

$$\vec{E}(\vec{r}) = -j\omega I_a^{TX} \frac{\mu_0}{4\pi} \frac{e^{-j\beta r}}{r} \vec{h}_{e\perp}^{TX}$$

where $\beta = \frac{2\pi}{\lambda}$ is the free-space wavenumber.

1. Direct Wave at Receiver

$$\vec{E}_1 = -j\omega I_a^{TX} \frac{\mu_0}{4\pi} \frac{e^{-j\beta d_1}}{d_1} h_{e\perp}^{TX} \vec{1}_z$$

2. Reflected Wave at Receiver

$$\vec{E}_2 = -j\omega I_a^{TX} \frac{\mu_0}{4\pi} \frac{e^{-j\beta d_2}}{d_2} h_{e\perp}^{TX} \Gamma_\perp(\theta_i) \vec{1}_z$$

Open-Circuit Voltage V_{oc} at the RX Terminals:

$$V_{oc} = -\vec{h}_{e\perp}^{RX} \cdot (\vec{E}_1 + \vec{E}_2) = j\omega \frac{\mu_0}{4\pi} I_a^{TX} h_{e\perp}^{TX} h_{e\perp}^{RX} \left[\frac{e^{-j\beta d_1}}{d_1} + \Gamma_\perp(\theta_i) \frac{e^{-j\beta d_2}}{d_2} \right]$$

Proof: Converting Circuit Voltage into Antenna Gains

For a matched receiver ($Z_L = Z_a^*$), the delivered power is $P_{RX} = \frac{1}{8R_a^{RX}}|V_{oc}|^2$:

$$P_{RX} = \frac{|I_a^{TX}|^2}{8R_a^{RX}}|h_{e\perp}^{TX}|^2|h_{e\perp}^{RX}|^2\frac{\omega^2\mu_0^2}{16\pi^2}\left|\frac{e^{-j\beta d_1}}{d_1}+\Gamma_\perp\frac{e^{-j\beta d_2}}{d_2}\right|^2$$

$$= \underbrace{\left(\frac{1}{2}R_a^{TX}|I_a^{TX}|^2\right)}_{P_{TX}}\cdot\frac{|h_{e\perp}^{TX}|^2}{R_a^{TX}}\frac{|h_{e\perp}^{RX}|^2}{R_a^{RX}}\frac{\omega^2\mu_0^2}{64\pi^2}\left|\frac{e^{-j\beta d_1}}{d_1}+\Gamma_\perp\frac{e^{-j\beta d_2}}{d_2}\right|^2$$

Fundamental Electromagnetic Identities

Using $\omega = \beta c = \frac{2\pi c}{\lambda}$, $c^2\mu_0^2 = Z_0^2$, and the Gain identity $G = \frac{\pi Z_0}{R_a}\frac{|h_{e\perp}|^2}{\lambda^2}$:

$$\frac{\omega^2\mu_0^2}{64\pi^2}\frac{|h_{e\perp}^{TX}|^2}{R_a^{TX}}\frac{|h_{e\perp}^{RX}|^2}{R_a^{RX}} = \left(\frac{\lambda}{4\pi}\right)^2\underbrace{\left(\frac{\pi Z_0|h_{e\perp}^{TX}|^2}{\lambda^2 R_a^{TX}}\right)}_{G_{TX}}\underbrace{\left(\frac{\pi Z_0|h_{e\perp}^{RX}|^2}{\lambda^2 R_a^{RX}}\right)}_{G_{RX}}$$

Exact Two-Ray Received Power Equation:

$$\boxed{P_{RX}(d) = \left(\frac{\lambda}{4\pi}\right)^2 G_{TX}G_{RX}P_{TX}\left|\frac{e^{-j\beta d_1}}{d_1}+\Gamma_\perp(\theta_i)\frac{e^{-j\beta d_2}}{d_2}\right|^2}$$

Fresnel Reflection Coefficient & The Grazing Limit

TE Fresnel Reflection Coefficient:

$$\Gamma_\perp(\theta_i) = \frac{\cos\theta_i - \sqrt{\epsilon_r - \sin^2\theta_i}}{\cos\theta_i + \sqrt{\epsilon_r - \sin^2\theta_i}}$$

where ϵ_r is the relative permittivity of the reflecting surface (ground here).

Asymptotic Grazing Limit ($d \gg h_{TX}, h_{RX}$)

As distance d increases:

$$\theta_i = \frac{\pi}{2} - \arctan\left(\frac{h_{TX} + h_{RX}}{d}\right) \to \frac{\pi}{2}$$

$$\lim_{\theta_i \to \pi/2} \Gamma_\perp(\theta_i) = \frac{0 - \sqrt{\epsilon_r - 1}}{0 + \sqrt{\epsilon_r - 1}} = -\mathbf{1}$$

Physical Interpretation:

- At small grazing angle ($\theta_i \to 90°$), our dielectric surface will behave as a **perfect electrical conductor**.
- The reflected wave gets a $\mathbf{180°}$ (π rad) phase shift upon reflection ($\Gamma_\perp \to -1$).
- The total field becomes a pure **destructive cancellation problem** controlled entirely by the geometric path difference $\Delta d = d_2 - d_1$.

Differential Path $\Delta d = d_2 - d_1$

For $d \gg (h_{TX} \pm h_{RX})$, amplitude terms are approximately equal: $d_1 \approx d_2 \approx d$. However, the phase difference $\beta(d_2 - d_1)$ is multiplied by $\beta = \frac{2\pi}{\lambda}$ and must be expanded to second order:

$$d_1 = d\sqrt{1 + \left(\frac{h_{TX} - h_{RX}}{d}\right)^2} \simeq d\left[1 + \frac{1}{2}\left(\frac{h_{TX} - h_{RX}}{d}\right)^2\right] = d + \frac{(h_{TX} - h_{RX})^2}{2d}$$

$$d_2 = d\sqrt{1 + \left(\frac{h_{TX} + h_{RX}}{d}\right)^2} \simeq d\left[1 + \frac{1}{2}\left(\frac{h_{TX} + h_{RX}}{d}\right)^2\right] = d + \frac{(h_{TX} + h_{RX})^2}{2d}$$

Exact Differential Path Difference Δd

Subtracting the two Taylor expansions:

$$\Delta d = d_2 - d_1 \simeq \frac{(h_{TX} + h_{RX})^2 - (h_{TX} - h_{RX})^2}{2d} = \frac{4h_{TX}h_{RX}}{2d} = \mathbf{\frac{2h_{TX}h_{RX}}{d}}$$

Differential Phase Shift:

$$\Delta\phi(d) = \beta(d_2 - d_1) \simeq \frac{2\pi}{\lambda}\left(\frac{2h_{TX}h_{RX}}{d}\right) = \frac{4\pi h_{TX}h_{RX}}{\lambda d}$$

Destructive Interference & Asymptotic d^{-4} Power Law

Substituting $\Gamma_\perp \simeq -1$ and $d_1 \approx d_2 \approx d$ into $P_{RX}(d) = \left(\frac{\lambda}{4\pi}\right)^2 G_{TX}G_{RX}P_{TX}\left|\frac{e^{-j\beta d_1}}{d_1} + \Gamma_\perp(\theta_i)\frac{e^{-j\beta d_2}}{d_2}\right|^2$

$$\begin{aligned} P_{RX} &\simeq \left(\frac{\lambda}{4\pi}\right)^2 \frac{G_{TX}G_{RX}P_{TX}}{d^2}\left|e^{-j\beta d_1} - e^{-j\beta d_2}\right|^2 \\ &= \left(\frac{\lambda}{4\pi}\right)^2 \frac{G_{TX}G_{RX}P_{TX}}{d^2}\left|1 - e^{-j\beta(d_2-d_1)}\right|^2 = 4\left(\frac{\lambda}{4\pi}\right)^2 \frac{G_{TX}G_{RX}P_{TX}}{d^2}\sin^2\left(\frac{\beta(d_2-d_1)}{2}\right) \end{aligned}$$

Small-Angle Taylor Approximation for $d \to \infty$

Since $\frac{\beta(d_2-d_1)}{2} = \frac{\beta h_{TX}h_{RX}}{d} \ll 1$, we use $\sin(x) \simeq x$:

$$\sin^2\left(\frac{\beta h_{TX}h_{RX}}{d}\right) \simeq \left(\frac{\beta h_{TX}h_{RX}}{d}\right)^2 = \left(\frac{2\pi}{\lambda}\frac{h_{TX}h_{RX}}{d}\right)^2$$

Final Asymptotic Power Law (Complete Cancellation of Wavelength λ):

$$P_{RX}(d) \simeq 4\left(\frac{\lambda}{4\pi}\right)^2 \frac{G_{TX}G_{RX}P_{TX}}{d^2}\cdot\frac{4\pi^2 h_{TX}^2 h_{RX}^2}{\lambda^2 d^2} \implies \boxed{P_{RX}(d) = G_{TX}G_{RX}P_{TX}\frac{h_{TX}^2 h_{RX}^2}{d^4}}$$

Analytical Derivation of the Breakpoint Distance d_{BP}

The propagation profile exhibits two fundamental regimes separated by a **breakpoint**:

- **Near Region ($d \leq d_{BP}$):** Constructive/destructive interference ripples around Friis loss ($n = 2$).
- **Far Region ($d > d_{BP}$):** Monotonic destructive cancellation with steep roll-off ($n = 4$).

Friis and Asymptotic Power Envelopes

The transition distance d_{BP} occurs where the asymptotic d^{-4} curve intersects Friis free-space power:

$$P_{RX,\text{Friis}}(d_{BP}) = P_{RX,\text{asympt}}(d_{BP})$$

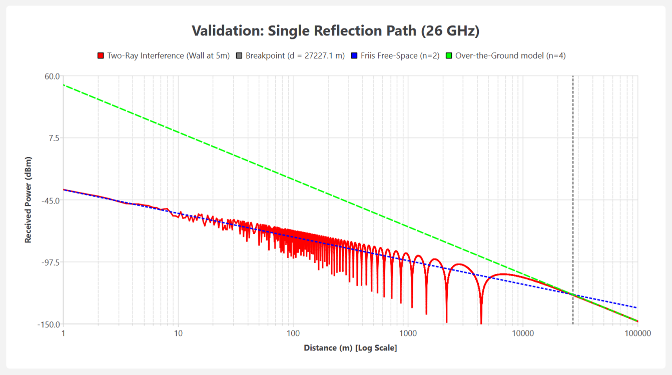
$$G_{TX} G_{RX} P_{TX} \left(\frac{\lambda}{4\pi d_{BP}} \right)^2 = G_{TX} G_{RX} P_{TX} \frac{h_{TX}^2 h_{RX}^2}{d_{BP}^4}$$

$$\frac{\lambda^2}{16\pi^2 d_{BP}^2} = \frac{h_{TX}^2 h_{RX}^2}{d_{BP}^4} \implies d_{BP}^2 = \frac{16\pi^2 h_{TX}^2 h_{RX}^2}{\lambda^2}$$

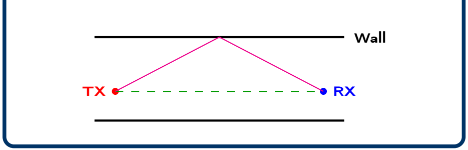
The Analytical Breakpoint Distance:

$$\boxed{d_{BP} = \frac{4\pi h_{TX} h_{RX}}{\lambda} = \frac{4\pi f h_{TX} h_{RX}}{c}}$$

Ray-Tracing Engine Validation (Project Results)



Validation Scene (App. A.2)



Simulation Setup (26 **GHz**)**:**

- $h_{TX} = h_{RX} = w = 5$ m, $\lambda = 1.15$ cm.
- $d_{BP} = \frac{4\pi(5)^2}{0.0115} = \mathbf{27\,227}$ m.
- Verifies the $n = 2 \rightarrow n = 4$ transition and asymptotic convergence!

Applied System Takeaways: Sub-6 GHz vs. mmWave 5G

Breakpoint Scaling Across 5G Operating Bands

Consider a street-level small cell where $h_{TX} = 4$ m (lamppost) and $h_{RX} = 1.5$ m (pedestrian UE):

5G Band	Frequency (f)	Wavelength (λ)	Breakpoint (d_{BP})	Physical Regime in Cell ($R \leq 150$m)
Low-Band	700 MHz	42.8 cm	**176** m	Reaches $n = 4$ asymptotic decay at cell edge
Mid-Band	3.5 GHz	8.57 cm	**880** m	Near-breakpoint transition zone
mmWave	26 GHz	1.15 cm	**6.54** km	**Strictly $n = 2$ with fast multipath ripples**

Design Consequence for mmWave

- At 26 GHz, small cells ($R \ll d_{BP}$) never enter the d^{-4} region.

Summary of Mathematical & Physical Results

1. Complete Theoretical Derivation Chain

$$V_{oc} = -\vec{h}_{e\perp} \cdot \vec{E}_{tot} \xrightarrow{\text{Matched Load}} P_{RX} = \left(\frac{\lambda}{4\pi}\right)^2 G_{TX} G_{RX} P_{TX} \left| \frac{e^{-j\beta d_1}}{d_1} + \Gamma_\perp \frac{e^{-j\beta d_2}}{d_2} \right|^2$$

$$\xrightarrow[d \gg h_{TX}, h_{RX}]{\Gamma_\perp \to -1,\ \Delta d \simeq \frac{2 h_{TX} h_{RX}}{d}} \boxed{P_{RX}(d) = G_{TX} G_{RX} P_{TX} \frac{h_{TX}^2 h_{RX}^2}{d^4}}$$ **(Frequency Independent!)**

2. The Breakpoint Distance

$$\boxed{d_{BP} = \frac{4\pi h_{TX} h_{RX}}{\lambda} = \frac{4\pi f h_{TX} h_{RX}}{c}}$$

Marks the boundary where ground cancellation permanently sets in.

3. Simulation Validation

The 2D C++ ray-tracer strictly confirms:

- Ripple dynamics for $d < d_{BP}$.
- Exact $n = 4$ asymptotic slope for $d > d_{BP}$.